higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 2

(6060452)

27 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL drawings must be neat and in good proportion.
5. Start each section on a NEW page
6. Write neatly and legibly.

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QUESTION 1

1.1 What is the purpose of an MIC? (2)

1.2 An MIC is essential for the operation of the various processes in industry.

1.2.1 Give THREE industry fields in which MIC systems are used. (3 × 2) (6)

1.2.2 Give a brief description of the processes controlled or products manufactured in the companies you listed in QUESTION 1.2.1 above. (9)

1.3 Explain the following words:

1.3.1 Hazardous

1.3.2 Combustible

(2 × 2) (4)
[21]

NCV

QUESTION 2

2.1 Complete the table below by writing the number down and the missing information next to it in the ANSWER BOOK.

TYPE	APPLICATIONS	ADVANTAGES	DISADVANTAGES
Plug valves	2.1.1 (2)	2.1.2 (1)	Requires constant lubrication for friction elimination
2.1.3 (1)	Used where a complete tight shut-off is not essential. Also used in cryogenic or high-temperature service and high vacuum conditions.	• Low cost • High-flow capacity • Good flow control • Low-pressure drop	2.1.4 (2) Give TWO disadvantages
2.1.5 (1)	• Used for most liquids. • Used for high temperatures	2.1.6 (3) Give THREE advantage	2.1.7 (1) Give ONE disadvantage
Three-way valve	• 2.1.8 (2)	• Allows for splitting of flow • Can be tight closing • High resistance to corrosion	2.1.9 (2) Give TWO disadvantages

(15)

2.2 When we talk about control valve accessories, we are referring to devices that are part of actuators. These devices help to make the actuator more functional.

2.2.1 Name THREE control valve accessories that you know.

2.2.2 Give an application of each accessory that you named in QUESTION 2.2.1.

2.2.3 What is the purpose of the accessories that you named in QUESTION 2.2.1? (9)

2.3 Name FOUR control valve components. (4)

2.4 Explain how the components you listed in QUESTION 2.3 helps in the operation of a valve. (4)

[32]

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QUESTION 3

3.1 Use the table given below to do the following conversions:

kPa	bar	psi	atm	Meter Hg
1	0,01	0,145	0,00986	0,0075
100	1	14,50	0,986	0,750
6,895	0,0689	1	0,0684	0,0517
101,32	1,01325	14,689	1	0,76
3,386	0,0338	0,491	0,334	0,0254

3.1.1 5 psi to bar

3.13.2 15 psi to kPa

(4 × 2) (8)

3.2 When using an equal leg manometer under dry conditions, explain by making calculations the height that the mercury will rise when a differential pressure of 2 bar (200 kPa) is applied. (5)

3.3 The Bourdon tube pressure gauge is one of the most widely used non-liquid pressure measurement devices.

3.3.1 Name TWO metals that Bourdon tubes are made from. (2)

3.3.2 What are the factors that can influence the free-end movement of a Bourdon tube? (5)

3.3.3 Name THREE calibration errors that you find in Bourdon tubes. (3)

3.3.4 Make a neat labeled sketch of a C-type Bourdon tube. (7)

[30]

QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

4.1.1 Thermal means relating to coldness.

4.1.2 When temperature increases, metals expands; when temperature decreases, metals shrink.

4.1.3 Thermometry is the measurement of temperature.

4.1.4 If water is boiling and you add heat to it, its temperature will change.

4.1.5 A thermocouple is a thermoelectric device for measuring the difference in temperature between two points.

4.1.6 Thermocouples can be used in electronics and can convert thermal potential difference into electric potential difference.

4.1.7 Thermowells are typically used to isolate a temperature sensor from the environment.

4.1.8 Thermowells are usually used with transducers and potentiometers.

4.1.9 Thermocouples can sense temperature on their own.

4.1.10 RTDs are useful when you want to measure temperature quickly but they are not accurate.

(10 × 1) (10)

4.2 Define thermocouples using:

4.2.1 The Seebeck effect

4.2.2 The Peltier effect

4.2.3 The Thompson effect

(3 × 4) (12)

4.3 Make neat labelled drawings of the following wiring configurations in RTDs.

4.3.1 Four wire connection

4.3.2 Three wire connection

(2 × 4) (8)

[30]

QUESTION 5

5.1 Choose a/an item/word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–L) next to the question number (5.1.1–5.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
5.1.1	A measurement of how much fuel a driver puts into a car at a petrol station so that he/she can be charged the correct amount	A	differential pressure flow
5.1.2	The flow caused by differences in pressures between two points	B	segment wedges
5.1.3	The speed of the flowing air in the middle of a pipe	C	pitot tube
5.1.4	This is based on the relationship between pressure and flow velocity	D	limitation of differential pressure flow measurements
5.1.5	Differential flow measurement instrument	E	orifice plates
5.1.6	The accuracy of orifice plates is usually influenced by changes in pressure, viscosity and density	F	linear velocity of the air
5.1.7	One of the oldest devices for measuring fluid velocity	G	float
5.1.8	They are manufactured from stainless steel and teflon	H	variable area flowmeters
5.1.9	They are used to provide flow rate indications for liquids, gasses and steam	I	tapered tube
5.1.10	A device floating on the surface of a liquid	J	vane meter
		K	flow measurement
		L	Bernoulli equation

(10 × 1) (10)

5.2 A Pitot tube is useful for measurements of flow. It can be used to measure the velocity of an aircraft. It actually measures the velocity of a stream at a point.

5.2.1 Make a neat sketch of a Pitot tube. (5)

5.2.2 Give THREE limitations of a Pitot tube. (3)

5.2.3 Give THREE advantages of a Pitot tube. (3)

5.2.4 Explain how a Pitot tube operates. (4)

5.3 Explain how you will go about installing an orifice plate flowmeter. (7)

5.4 Explain how you will zero check an orifice plate flowmeter. (5)

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TOTAL: 150

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